

Chemical Equipment Analysis Report

Dataset: sample148.csv
Upload Date: 2026-02-03 07:34
Total Equipment Count: 147

Summary Statistics

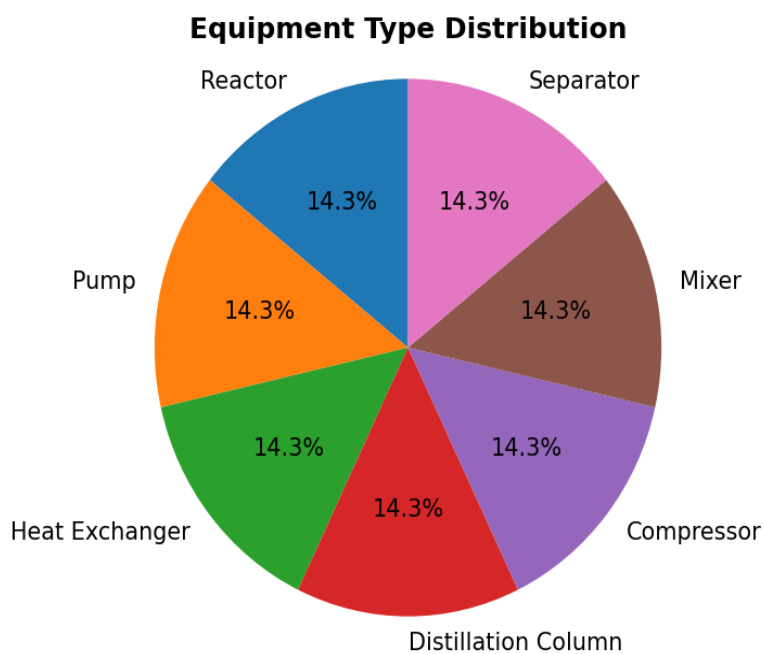
Metric	Value	Unit
Average Flowrate	193.65	L/min
Average Pressure	60.62	psi
Average Temperature	106.84	°C

Equipment Type Distribution

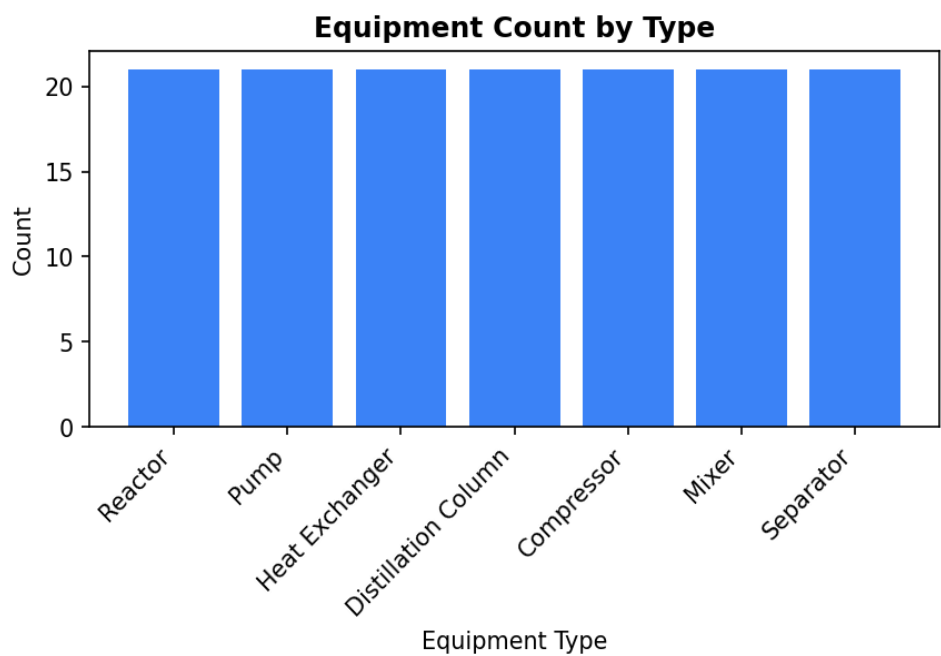
Equipment Type	Count	Percentage
Reactor	21	14.3%
Pump	21	14.3%
Heat Exchanger	21	14.3%
Distillation Column	21	14.3%
Compressor	21	14.3%
Mixer	21	14.3%
Separator	21	14.3%

Visual Analysis

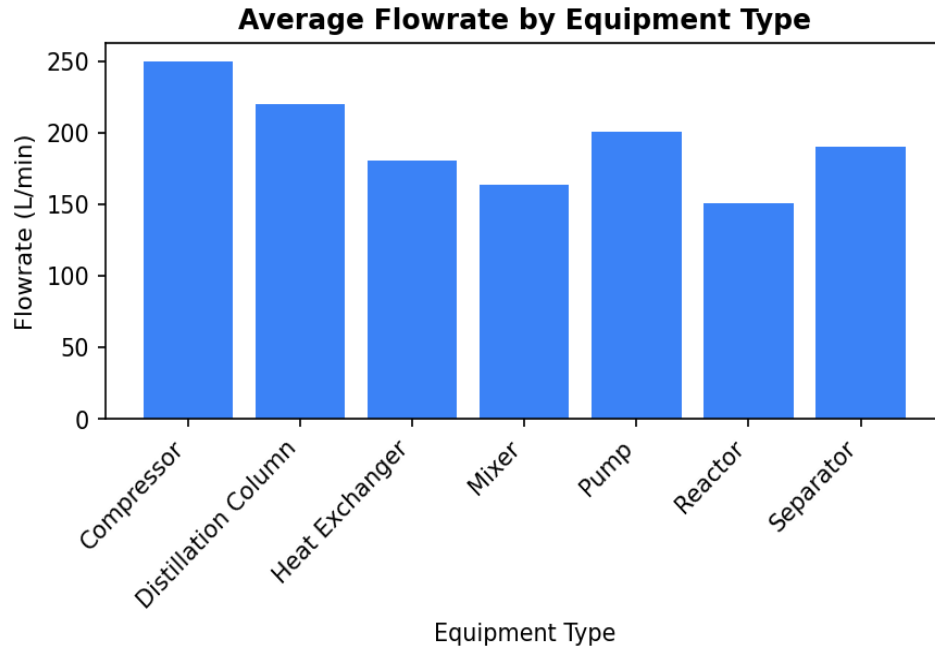
Equipment Type Distribution



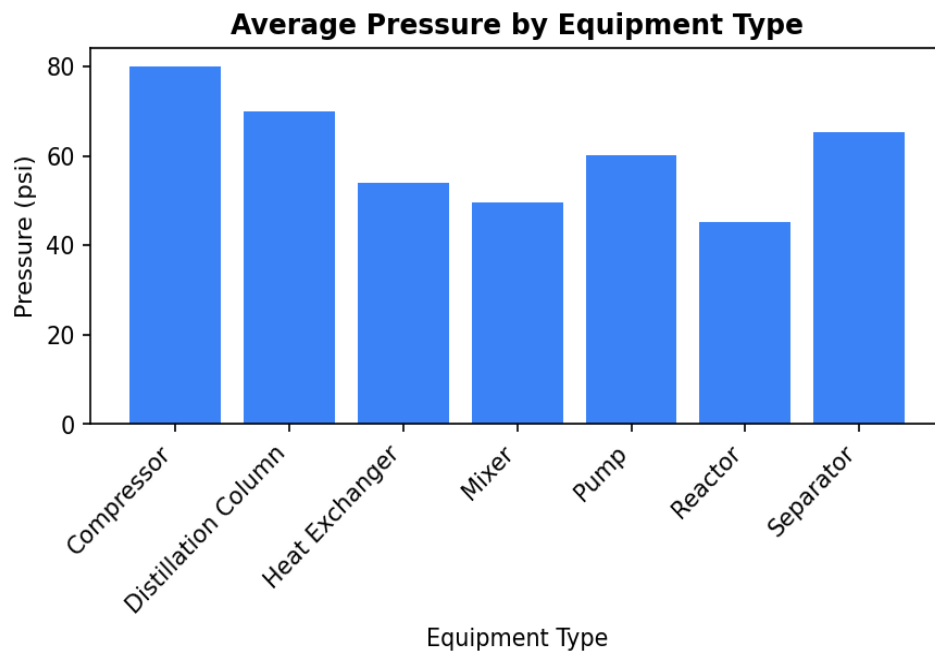
Equipment Count by Type



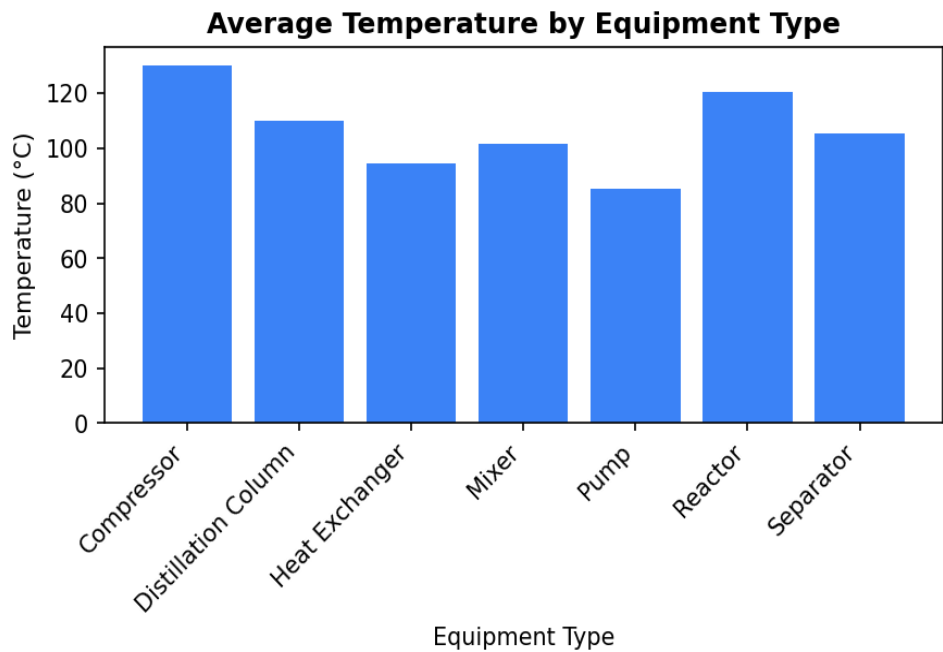
Average Flowrate by Equipment Type



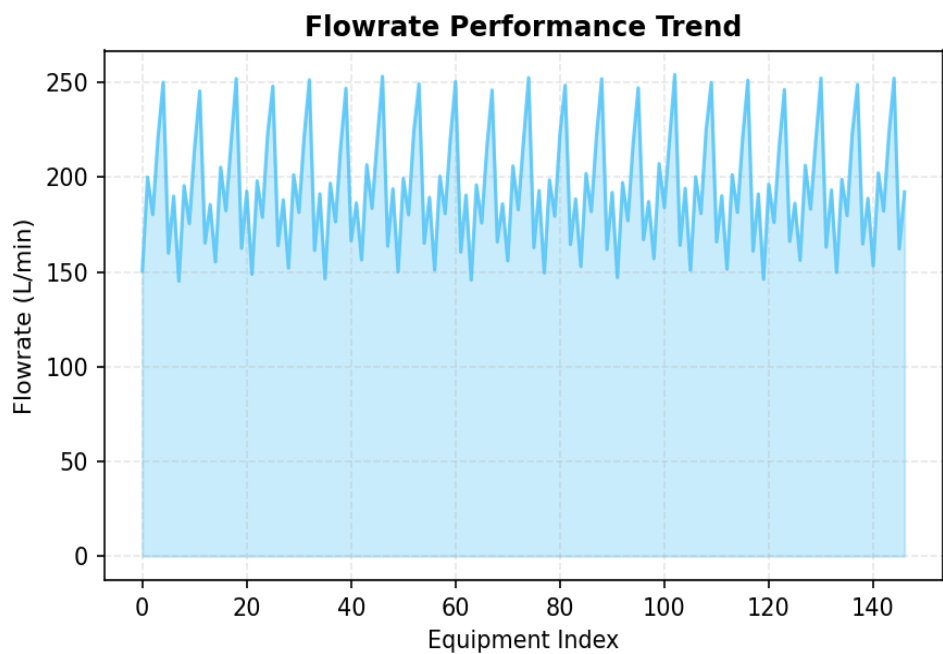
Average Pressure by Equipment Type



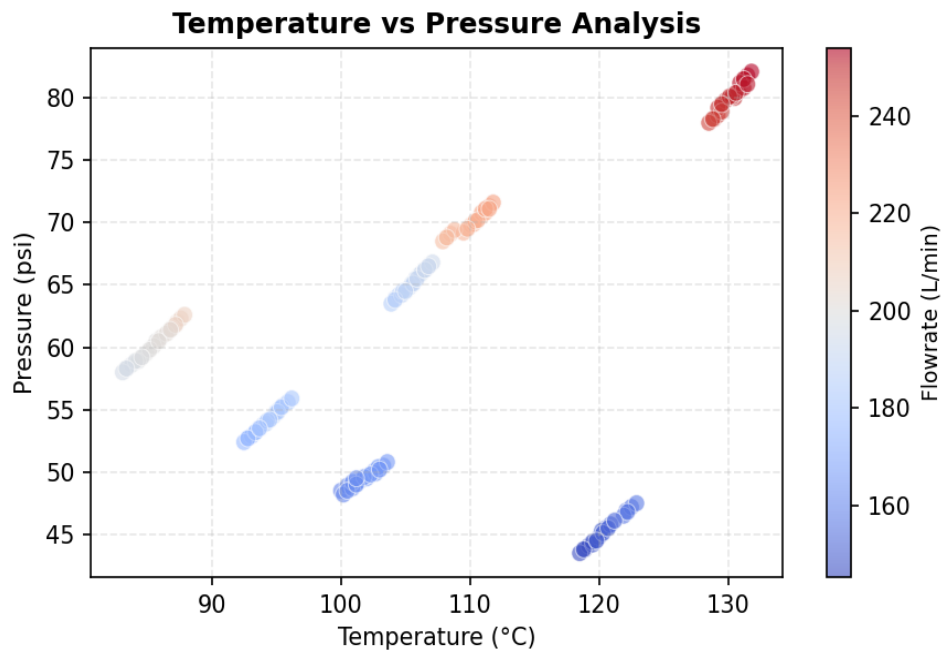
Average Temperature by Equipment Type



Flowrate Performance Trend (All Equipment)



Temperature vs Pressure Analysis (All Equipment)



Detailed Equipment Data

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Reactor-A1	Reactor	150.50	45.20	120.30
Pump-B1	Pump	200.00	60.50	85.70
Heat Exchanger-C1	Heat Exchanger	180.30	55.00	95.20
Distillation Column-D1	Distillation Column	220.80	70.10	110.50
Compressor-E1	Compressor	250.00	80.00	130.50
Mixer-F1	Mixer	160.00	48.50	100.00
Separator-G1	Separator	190.00	65.50	105.80
Reactor-A2	Reactor	145.20	43.80	118.90
Pump-B2	Pump	195.50	58.30	83.40
Heat Exchanger-C2	Heat Exchanger	175.60	52.70	92.80
Distillation Column-D2	Distillation Column	215.00	68.80	108.20
Compressor-E2	Compressor	245.50	78.30	128.90
Mixer-F2	Mixer	165.30	49.80	102.30
Separator-G2	Separator	185.50	63.90	104.20
Reactor-A3	Reactor	155.30	46.90	122.10
Pump-B3	Pump	205.20	62.00	87.30
Heat Exchanger-C3	Heat Exchanger	182.40	54.20	94.50

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Distillation Column-D3	Distillation Column	218.00	69.50	109.80
Compressor-E3	Compressor	252.00	81.50	131.20
Mixer-F3	Mixer	162.50	49.00	101.20
Separator-G3	Separator	192.50	66.20	106.50
Reactor-A4	Reactor	148.80	44.50	119.70
Pump-B4	Pump	198.00	59.20	84.60
Heat Exchanger-C4	Heat Exchanger	178.90	53.50	93.70
Distillation Column-D4	Distillation Column	222.50	71.00	111.20
Compressor-E4	Compressor	248.00	79.50	129.50
Mixer-F4	Mixer	164.00	50.20	103.00
Separator-G4	Separator	188.00	64.50	105.00
Reactor-A5	Reactor	152.10	45.50	120.50
Pump-B5	Pump	201.20	60.80	86.10
Heat Exchanger-C5	Heat Exchanger	181.50	55.30	95.50
Distillation Column-D5	Distillation Column	221.40	70.50	110.90
Compressor-E5	Compressor	251.30	80.50	130.80
Mixer-F5	Mixer	161.40	48.90	100.50
Separator-G5	Separator	191.20	65.80	106.10
Reactor-A6	Reactor	146.50	44.10	119.20
Pump-B6	Pump	196.80	58.60	83.80
Heat Exchanger-C6	Heat Exchanger	176.50	52.90	93.10
Distillation Column-D6	Distillation Column	216.40	69.10	108.50
Compressor-E6	Compressor	246.80	78.60	129.20
Mixer-F6	Mixer	166.50	50.10	102.60
Separator-G6	Separator	186.40	64.20	104.50
Reactor-A7	Reactor	156.40	47.20	122.50
Pump-B7	Pump	206.50	62.30	87.60
Heat Exchanger-C7	Heat Exchanger	183.60	54.50	94.80
Distillation Column-D7	Distillation Column	219.40	69.80	110.10
Compressor-E7	Compressor	253.20	81.80	131.50
Mixer-F7	Mixer	163.70	49.30	101.50
Separator-G7	Separator	193.80	66.50	106.80
Reactor-A8	Reactor	150.10	44.80	120.00
Pump-B8	Pump	199.30	59.50	84.90
Heat Exchanger-C8	Heat Exchanger	180.20	53.80	94.00
Distillation Column-D8	Distillation Column	223.80	71.30	111.50
Compressor-E8	Compressor	249.20	79.80	129.80

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Mixer-F8	Mixer	165.20	50.50	103.30
Separator-G8	Separator	189.30	64.80	105.30
Reactor-A9	Reactor	151.00	45.30	120.20
Pump-B9	Pump	200.50	60.10	85.50
Heat Exchanger-C9	Heat Exchanger	180.80	54.80	95.00
Distillation Column-D9	Distillation Column	220.10	69.90	110.30
Compressor-E9	Compressor	250.40	80.10	130.30
Mixer-F9	Mixer	160.50	48.20	100.20
Separator-G9	Separator	190.50	65.10	105.50
Reactor-A10	Reactor	145.80	43.50	118.50
Pump-B10	Pump	195.90	58.00	83.10
Heat Exchanger-C10	Heat Exchanger	175.90	52.40	92.50
Distillation Column-D10	Distillation Column	215.50	68.50	107.90
Compressor-E10	Compressor	245.90	78.00	128.50
Mixer-F10	Mixer	165.90	49.50	102.00
Separator-G10	Separator	185.90	63.50	103.90
Reactor-A11	Reactor	155.90	46.50	121.90
Pump-B11	Pump	205.90	61.50	86.90
Heat Exchanger-C11	Heat Exchanger	182.90	53.90	94.20
Distillation Column-D11	Distillation Column	218.50	69.20	109.50
Compressor-E11	Compressor	252.50	81.20	130.90
Mixer-F11	Mixer	162.90	48.70	100.90
Separator-G11	Separator	192.90	65.90	106.20
Reactor-A12	Reactor	149.50	44.20	119.50
Pump-B12	Pump	198.50	58.90	84.30
Heat Exchanger-C12	Heat Exchanger	179.50	53.20	93.40
Distillation Column-D12	Distillation Column	222.90	70.80	110.90
Compressor-E12	Compressor	248.50	79.20	129.20
Mixer-F12	Mixer	164.50	49.90	102.70
Separator-G12	Separator	188.50	64.20	104.70
Reactor-A13	Reactor	152.90	45.80	120.90
Pump-B13	Pump	201.90	61.10	86.50
Heat Exchanger-C13	Heat Exchanger	181.90	55.60	95.90
Distillation Column-D13	Distillation Column	221.90	70.80	111.20
Compressor-E13	Compressor	251.90	80.80	131.20
Mixer-F13	Mixer	161.90	49.20	100.90
Separator-G13	Separator	191.90	66.20	106.50

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Reactor-A14	Reactor	147.10	44.40	119.50
Pump-B14	Pump	197.10	58.90	84.10
Heat Exchanger-C14	Heat Exchanger	177.10	53.20	93.40
Distillation Column-D14	Distillation Column	217.10	69.40	108.80
Compressor-E14	Compressor	247.10	78.90	129.50
Mixer-F14	Mixer	167.10	50.40	102.90
Separator-G14	Separator	187.10	64.50	104.80
Reactor-A15	Reactor	157.10	47.50	122.90
Pump-B15	Pump	207.10	62.60	87.90
Heat Exchanger-C15	Heat Exchanger	184.10	54.80	95.10
Distillation Column-D15	Distillation Column	220.10	70.10	110.40
Compressor-E15	Compressor	254.10	82.10	131.80
Mixer-F15	Mixer	164.10	49.60	101.80
Separator-G15	Separator	194.10	66.80	107.10
Reactor-A16	Reactor	150.90	45.10	120.30
Pump-B16	Pump	200.10	59.80	85.20
Heat Exchanger-C16	Heat Exchanger	180.90	54.10	94.30
Distillation Column-D16	Distillation Column	224.90	71.60	111.80
Compressor-E16	Compressor	249.90	80.10	130.10
Mixer-F16	Mixer	165.90	50.80	103.60
Separator-G16	Separator	190.10	65.10	105.60
Reactor-A17	Reactor	151.50	45.50	120.70
Pump-B17	Pump	201.20	60.50	85.90
Heat Exchanger-C17	Heat Exchanger	181.50	55.20	95.40
Distillation Column-D17	Distillation Column	220.80	70.20	110.60
Compressor-E17	Compressor	251.10	80.40	130.60
Mixer-F17	Mixer	161.10	48.50	100.50
Separator-G17	Separator	191.10	65.50	105.90
Reactor-A18	Reactor	146.20	43.80	118.80
Pump-B18	Pump	196.20	58.30	83.40
Heat Exchanger-C18	Heat Exchanger	176.20	52.70	92.80
Distillation Column-D18	Distillation Column	216.20	68.80	108.20
Compressor-E18	Compressor	246.20	78.30	128.80
Mixer-F18	Mixer	166.20	49.80	102.30
Separator-G18	Separator	186.20	63.80	104.20
Reactor-A19	Reactor	156.20	46.80	122.20
Pump-B19	Pump	206.20	61.80	87.20

Equipment Name	Equipment Type	Flowrate (L/min)	Pressure (psi)	Temperature (°C)
Heat Exchanger-C19	Heat Exchanger	183.20	54.20	94.50
Distillation Column-D19	Distillation Column	219.20	69.50	109.80
Compressor-E19	Compressor	252.20	81.50	131.20
Mixer-F19	Mixer	163.20	49.00	101.20
Separator-G19	Separator	193.20	66.20	106.50
Reactor-A20	Reactor	149.80	44.50	119.80
Pump-B20	Pump	198.80	59.20	84.60
Heat Exchanger-C20	Heat Exchanger	179.80	53.50	93.70
Distillation Column-D20	Distillation Column	223.20	71.10	111.20
Compressor-E20	Compressor	248.80	79.50	129.50
Mixer-F20	Mixer	164.80	50.20	103.00
Separator-G20	Separator	188.80	64.50	105.00
Reactor-A21	Reactor	153.20	46.10	121.20
Pump-B21	Pump	202.20	61.40	86.80
Heat Exchanger-C21	Heat Exchanger	182.20	55.90	96.20
Distillation Column-D21	Distillation Column	222.20	71.10	111.50
Compressor-E21	Compressor	252.20	81.10	131.50
Mixer-F21	Mixer	162.20	49.50	101.20
Separator-G21	Separator	192.20	66.50	106.80

Total rows displayed: 147